

MAJOR CONTRIBUTORS TO THIS REPORT



Amol Kadu
regional manager -
west region, Schenck
Technologies Pvt Ltd



Madhukar Tripathi
senior manager -
marketing and channel
sales, Anritsu India
Pvt Ltd



Naresh Narasimhan
country marketing
manager, Tektronix
India



Pathy V.V.
business development
manager - education,
Keysight Technologies
India Pvt Ltd



Satish Mohanram
technical marketing
manager, National
Instruments



Sumit Sharma
marketing manager -
India, GW Instek

specific measurement automation functions to push engineering students to explore results on their own and learn how to handle more basic measurement functions. It is also suitable for general-purpose measurement tasks in R&D, manufacturing and maintenance.

Teaching labs/design suites.

Talking about teaching labs, Pathy informs, "On the digital communication side, we have introduced a new teaching lab that educates students on digital modulation/demodulation techniques." He adds, "For instance, consider a GSM phone. What does a GSM signal look like? The teaching lab focuses on such aspects."

As mentioned before, Anritsu has developed an RF and microwave trainer kit with Foundation for Innovation and Technology, IIT Delhi. Madhukar Tripathi, senior manager - marketing and channel sales, Anritsu India Pvt Ltd, informs, "This kit contains many passive components that are tested using Anritsu's vector network analyser (VNA). This gives a good overview for students to learn about real telecom components and their behaviour."

Traditionally, if somebody has to design a communication system, he or she has to go through a very tedious process of coming up with an algorithm. Mohanram says, "In India, due to lack of tools that can take those algorithms on to a physical hardware, a lot of research would get stuck at theoretical level. LabVIEW communication system design suite is one tool that helps take those designs, whether an algorithm written in language C or

LabVIEW, directly to a hardware and validate the designs very quickly." It also helps people convert and optimise some of the algorithms for FPGAs, wherever required.

The hardware known as the universal software defined radio along with LabVIEW communication system design suite together have helped academicians, who are researching or learning concepts in communication, to go leaps and bounds from their theory, putting all of it to practice, claims Mohanram. He says, "This is another interesting product that we have launched recently, which is now being used across academia and industry for learning, researching and implementing radio communication."

Initiatives by T&M vendors for academia

On the semiconductor side, government has recently approved setting up of two semiconductor fabrication plants (fabs); one in Gujarat and the other one at Noida, informs Pathy. He says, "We are actively engaged with Tier I institutes, such as IIT Gandhinagar, that are close to the new fab ecosystem." He adds, "We are helping students with courses. For example, there is one credit course that is developed and conducted by a technical person from Keysight."

Anritsu too conducts workshop seminars in engineering colleges and universities, and updates students with recent technological advancements. Tripathi says, "These sessions are followed by hands-on, live demo of products."

Keysight is also playing the role of an enabler in connecting some of the engineering colleges with the industry. Pathy says, "For example, R.V. College of Engineering in Bengaluru has set up a wireless lab. But by simply buying tools and providing training, it is not necessary that the students would be job-ready. Therefore we are trying to bring industries in and

around Bengaluru to start interacting with faculty of the college." For the industry to interact, there must be good infrastructure support available, like at R.V. College, which motivates them to start such an engagement. He adds, "From a college perspective, the very same interaction can also help the faculty to know some of the R&D programmes that these industries are into and general technology trends. Students are also given an opportunity to interact and understand job requirements in this industry."

Talking about National Instruments, Mohanram says, "We have always been doing hands-on workshops that happen almost on a weekly basis at our office, and these can be attended by both industry and academia before these take on the product and start using it."

Keysight has made significant investment into their R&D facility in India. Pathy says, "With our R&D team's help, we are able to offer a six-month student internship programme. This is targeted at students in the PG curriculum." He informs, "The student has access to our equipment and tools, and is also mentored by our R&D team. There is a sense of learning and contribution at the same time, as the student is involved in an active project."

T&M products that are there in the market help people measure and relate to what they learn. Today, T&M vendors are playing a significant role in making sure that students and academicians learn the concept they hear about very effectively using their tools. ●